

Tengyang Deng

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EDUCATION

Johns Hopkins University

Aug. 2025 – Present

Master of Science in Engineering, Computer Science | Baltimore, MD

Relevant Coursework: Algorithms, ML/DL, Distributed Systems, LLM, NLP, Agents

University of Ottawa

Sep. 2020 – Apr. 2025

Bachelor of Science (Honours), Computer Science | Ottawa, ON, Canada

GPA: 8.38/10 (3.75/4.0)

Honors: Merit Scholarship (2021, 2022, 2023) — awarded for term GPA above 8.5; Dean's Honour List (Winter 2022, Fall 2023)

RESEARCH & PROJECTS

PAWM: Perspective-Aware World Models for Bayesian Theory of Mind Inference Jan. 2026 – Present

Course Research Project, Johns Hopkins University | Advised by the SCAI Lab

- Identified a systematic architectural failure in AutoToM (NeurIPS 2025 Spotlight): evaluated the model on the BigToM Backward Belief False-Belief benchmark (200 questions), achieving 76.0% accuracy, and manually classified all 48 errors—attributing 100% to information asymmetry caused by the absence of per-agent observability constraints.
- Conducted structural analysis of AutoToM's internal Bayesian network variable files across all 200 stories, finding that 85% of false-belief stories are modeled as single-timestep snapshots and 83.3% of error cases contain no information asymmetry signal, pinpointing contamination at the variable construction stage.
- Designed and implemented PAWM, a lightweight preprocessing module that performs fork detection, story segmentation, and state variable constraint injection—preventing AutoToM's Bayesian Inverse Planning from leaking observer-privileged world state into predicted agent beliefs.
- Evaluated the full system on three BigToM splits (600 questions total) with ablation studies, targeting publication as a 4-page technical report in ACL format.

Technologies: Python, GPT-4o API, AutoToM (SCAI-JHU codebase), BigToM benchmark

Decoding-Tree-Sketching Migration: Hugging Face → vLLM

Feb. 2026 – March

Software Engineering Contributor, Johns Hopkins University

- Leading the migration of the Decoding-Tree-Sketching speculative decoding algorithm from the Hugging Face Transformers framework to vLLM, enabling high-throughput inference on large language models.
- Refactored core decoding logic to conform to vLLM's asynchronous engine architecture and PagedAttention memory management system.
- Validated correctness and performance of the ported implementation through systematic benchmarking against the original Hugging Face baseline.

Technologies: Python, PyTorch, vLLM, Hugging Face Transformers, CUDA

Web Application for a Meta-Learning Algorithm (Honours Project)

Sep. 2024 – Dec. 2024

Research Member, supervised by Professor Paula Branco — University of Ottawa

- Developed a full-stack web application using Flask to expose a Python-based meta-learning (METALLIC) model through an interactive HTML interface.
- Implemented dataset upload, meta-feature computation, and on-demand model inference, enabling end-to-end user-driven prediction workflows.
- Integrated session cookies to allow background model execution to persist across browser sessions.
- Designed a responsive, user-friendly UI following established web accessibility principles.

Languages & Tools: Python, HTML, JavaScript, Flask, GitHub, VS Code

YOLOv9 Object Detection Research

Aug. 2024 – Sep. 2024

Research Member, supervised by Associate Professor Guibo Zhu — Chinese Academy of Sciences

- Reproduced and analyzed the YOLOv9 architecture by studying the source code and original research paper; trained the model locally on the COCO dataset using Google Colab.
- Tuned hyperparameters and modified model components to improve detection accuracy and accelerate convergence.
- Curated and preprocessed a representative COCO image subset, ensuring data diversity, accuracy, and labeling completeness.
- Authored technical documentation covering algorithm principles, architecture design decisions, and experimental results.

Languages & Tools: Python, PyTorch, Docker, Jupyter Notebook, Google Colab

Tabu Search for the Graph Coloring Problem

Jan. 2023 – May 2023

Research Assistant, supervised by Professor Zhipeng Lv — Huazhong University of Science & Technology

- Studied and implemented the Tabu Search metaheuristic from foundational literature and applied it to the NP-hard Graph Coloring Problem (GCP).
- Optimized the algorithm's search strategy, achieving measurable reductions in runtime while accurately replicating published benchmark results.

Languages & Tools: C++, VS Code

COURSE PROJECTS

Video-Sharing Website UI Development

May 2024 – Jul. 2024

Independent Project, supervised by Professor Caroline Barrière — University of Ottawa

- Built a high-fidelity prototype of a video-sharing platform using React and Bootstrap, implementing features including video upload, filtered browsing, playlist management, and English/Chinese language switching.

Languages & Tools: HTML, CSS, JavaScript, React, GitHub, VS Code

Climate Change Dataset Preprocessing & Analytics

Jan. 2024 – May 2024

Collaborative Project, supervised by Professor Yazan Otoum — University of Ottawa

- Executed a four-phase data pipeline (planning, conceptual design, staging, and analytics) on a Kaggle climate-change dataset, leveraging OLAP queries and BI dashboards to surface actionable insights.
- Cleaned and transformed raw data using Python and Jupyter Notebook to ensure suitability for downstream analysis.

Languages & Tools: Python, Jupyter Notebook, BI Dashboard, Kaggle, GitHub

PROFESSIONAL EXPERIENCE

Harmo Robotics

Jul. 2023 – Sep. 2023

Technician, Research and Development Department

- Acquired comprehensive knowledge of industrial robotics systems, including operational principles, automation control, and client-facing applications.
- Programmed, assembled, and troubleshoot industrial robotic arms; developed proficiency in automation workflows under production conditions.
- Optimized sensor interface code in C++ using the Qt framework, improving execution speed and measurement accuracy.

Languages & Tools: C++, Qt

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, JS, Go, Scheme, Prolog

Frameworks & Libraries: PyTorch, vLLM, Hugging Face Transformers, React, Angular, Flask, Spring Boot, Bootstrap, Pandas, Scikit-learn

Databases: MySQL, PostgreSQL, Firebase

Tools & Technologies: Git/GitHub, Docker, VS Code, Jupyter Notebook, Google Colab, Microsoft Office Suite

LANGUAGES

Chinese (Native), English (Professional Working Proficiency)